CURVES WITHOUT RATIONAL TAILS: CONIC IN GENUS 0, TRIVIAL DUALIZING SHEAF IN GENUS 1, GLOBALLY GENERATED POWERS IN GENUS AT LEAST 2

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the Gorenstein input, the genus-0 dichotomy, the genus-1 degree bookkeeping with the triviality conclusion, and the genus->=2 reducible-case inequality check are all retained verbatim.

Editorial note. The label lemma-no-rational-tail-semiample-genus-geq-2 names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Nodal curves are Gorenstein and the tail inequalities. Same chapter, Lemma lemma-facts-about-nodal-curves; Example example-rational-tail; Lemma lemma-no-rational-tail. Makes omega_X invertible and supplies the componentwise degree inequalities ruling out rational tails.
- Genus zero and one inputs. Same chapter, Lemmas lemma-genus-zero and lemma-genus-gorenstein; Varieties, Lemmas varieties-lemma-degree-in-terms-of-components and varieties-lemma-no-sections-dual-nef. Classifies the irreducible genus-zero case as a plane conic with anti-dualizing very ample, and gives the degree bookkeeping and duality argument forcing omega_X trivial in genus one.
- Global generation inputs for genus at least two. Same chapter, Lemmas lemma-degree-more-than-2g and lemma-global-generation. Handles the irreducible case directly and supplies the inequality framework whose Y = X case is re-analysed in the reducible case.

Source reference. Original reference 0E3H: sources/stacks-curves/pinned-original-source.tex, source lines 5933–5933. Prerequisite pointer invoked at proof line(s) 6084; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0E3C: sources/stacks-curves/pinned-original-source.tex, source lines 5593–5593. Prerequisite pointer invoked at proof line(s) 6118; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:curves.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 0E37: sources/stacks-curves/pinned-original-source.tex, source lines 4763–4763. Prerequisite pointer invoked at proof line(s) 6095; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0C19: sources/stacks-curves/pinned-original-source.tex, source lines 1744–1744. Prerequisite pointer invoked at proof line(s) 6110; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0C6N: sources/stacks-curves/pinned-original-source.tex, source lines 2108–2108. Prerequisite pointer invoked at proof line(s) 6103; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0E3F: sources/stacks-curves/pinned-original-source.tex, source lines 5755–5755. Prerequisite pointer invoked at proof line(s) 6121; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0E3J: sources/stacks-curves/pinned-original-source.tex, source lines 6032–6032. Prerequisite pointer invoked at proof line(s) 6101; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AYW: sources/stacks-chow/tags.txt, source lines 13804–13804. Prerequisite pointer invoked at proof line(s) 6111; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0E22: sources/stacks-chow/tags.txt, source lines 17592–17592. Prerequisite pointer invoked at proof line(s) 6113; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let k be a field. Let X be a proper scheme over k having dimension 1 and $H^0(X, \mathcal{O}_X) = k$. Assume the singularities of X are at-worst-nodal. Assume X does not have a rational tail (Example 0E3H). If*

1. *the genus of X is 0, then X is isomorphic to an irreducible plane conic and $\omega_X^{\otimes -1}$ is very ample,*
2. *the genus of X is 1, then $\omega_X \cong \mathcal{O}_X$,*
3. *the genus of X is ≥ 2, then $\omega_X^{\otimes m}$ is globally generated for $m \geq 2$.*

Proof. By Lemma 0E37 we find that X is Gorenstein, i.e., ω_X is an invertible $\mathcal{O}_X$-module.

If the genus of X is zero, then $\deg(\omega_X) < 0$, hence if X has more than one irreducible component, we get a contradiction with Lemma 0E3J. In the irreducible case we see that X is isomorphic to an irreducible plane conic and $\omega_X^{\otimes -1}$ is very ample by Lemma 0C6N.

If the genus of X is 1, then ω_X has a global section and $\deg(\omega_X|_C) = 0$ for all irreducible components. Namely, $\deg(\omega_X|_C) \geq 0$ for all irreducible components C

by Lemma 0E3J, the sum of these numbers is 0 by Lemma 0C19, and we can apply Varieties, Lemma 0AYW. Then $\omega_X \cong \mathcal{O}_X$ by Varieties, Lemma 0E22.

Assume the genus g of X is greater than or equal to 2. If X is irreducible, then we are done by Lemma 0E3C. Assume X reducible. By Lemma 0E3J the inequalities of Lemma 0E3F hold for every $Y \subset X$ as in the statement, except for $Y = X$. Analyzing the proof of Lemma 0E3F we see that (in the reducible case) the only inequality used for $Y = X$ are

$$\deg(\omega_X^{\otimes m}) > -2\chi(\mathcal{O}_X) \quad \text{and} \quad \deg(\omega_X^{\otimes m}) + \chi(\mathcal{O}_X) > \dim_k H^1(X, \mathcal{O}_X)$$

Since these both hold under the assumption $g \geq 2$ and $m \geq 2$ we win. □